EL NIÑO ACTION BRIEFING · 4 NOVEMBER 2026

Talara, Piura, Peru

Everyone will watch the quebradas, but the failure that keeps recurring is one water line. Shore it and agree a cistern plan by 30 November.

PROVINCE	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Piura Region (province level)	360 forecast, region, peak 94 in December; 110 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Rain, flash floods, and water-line failure; fever and heat in infants and older people

19/19 recommendations checked: 10 ranked actions and 9 For you steps. 18 held only after corrections, 1 is unverified, and none passed uncorrected. 8 items are in Do not do this. Verification moved the funding target by 70%, from S/1.78 million to S/531,000 (US$153,000), and two proposed placements were dropped. Exposure figure: Climate Impact Lab Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, heat only and for all of Piura Region, not Talara. Status: ENFEN's communiqué of 30 October expects extraordinary sea warming through February and above-normal rain from November, so the clock has already started.

EXECUTIVE SUMMARY — AT A GLANCE



4.58°S 81.27°W, Peru · Natural Earth

- **The risk.** One aging water line serves more than 123,000 people and has no repair we could find; 2023's floods recorded no deaths, but 18 young children died here in 1997-98.
- **For you.** As a refinery manager, re-contract the 10 motor pumps by 15 November, for about S/178,020 (US$51,400).
- **The money.** Floor S/215,000 (US$62,000), target S/531,000 (US$153,000), ceiling S/1.11 million (US$320,000); household chlorine buys the most.
- **Do not.** Do not give ibuprofen or other anti-inflammatories for fever in the rainy season.*

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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A REFINERY MANAGER — 9 STEPS

In Talara, El Niño reaches a refinery through its own drains, the port, the grid, and its seawater intake, and the town will look to it when the water line fails. These steps assume Petroperú's refinery, the only one we found here, and rest on the Contraloría's visit of 19–25 August 2026, so you will know of later changes. As of 3 November the refinery was stopped after a national grid event on 1 November, with no restart date given; it had been running at 64% of capacity in late October. Petroperú could not pay contractors in August, and ProInversión plans a concession proposal for January 2027. The ranked actions below are written for the district and its partners. These are the ones you can take yourself.

1 · Pumps: re-contract the 10 motor pumps and start hour logs

PARTIALLY CONFIRMED | ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
15 November 2026	S/178,020 (US$51,400), the accepted offer in the report; re-letting may differ	Plant drainage and emergency response

The Contraloría's report 015-2026-OCI/0084-SVC records that the pump service never started, the contract was terminated on 16 April, and no hour logs exist. Its estimate was S/200,000 (US$57,700). The date is our proposal; the report gave five working days to report back, and we found no public reply.

2 · Channels: clear your rainwater channels of sludge and vegetation, then the gate duct, jointly with the municipality

PARTIALLY CONFIRMED | FROM ACTION 6

BY WHEN	COST TO YOU	PROTECTS
30 November 2026	No published figure; use your contractor rates	Site access and plant drainage

The report found both, and channel conditions were still being drafted. In 2023 Petroperú reported mud blocking drainage at its gate; another account names a canal-outlet blockage and contractors clearing ducts. Clear your own first, because that is what you control.

3 · Escalate this week: the racks clean-up, the former API separator, the Overales supports, and the Colector 3 outfall

* Suggested formulation: check with a health worker before sharing.

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
15 November 2026	Racks S/1,000,000 (US$288,500); API S/500,000 (US$144,300); others not priced	Sea, beaches, and the permit

These need corporate money. The report records the racks job unfinished "por falta de pago" with memos unanswered, oil at the API separator tracing to a January 2025 overflow with an OEFA file open, supports on the 1975 Overales line, and erosion at the Colector 3 outfall. The cracks reported in the press are in walls beside the submarine line, not in the line. In August contractors could not be paid, so a written, dated request is your protection.

4 · Spills: test the contingency plan against the areas where oily water is still open, before heavy rain

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
30 November 2026	Staff time	Coast and fishing grounds

DS 039-2014-EM, Articles 66 and 66-A (first response and contingency plan), are as quoted in the report. OEFA's S/192,600 (US$55,600) fine for the 2024 spill was coercive, for failing to identify and control the source. We found no specific Osinergmin pre-rain duty, and no documented rain-caused spill at Talara, which is not proof there has been none.

5 · Tanks: set two inventory triggers, high product and low crude, for port closures

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
15 December 2026	None	Avoiding an unplanned stop

Swells have closed ports in both directions: in 2025 product tanks filled and the refinery went on standby, and in July 2026 closures blocked crude during a shortage. Crude purchasing is probably not your call (our inference), so agree the low-crude trigger with whoever buys.

* Suggested formulation: check with a health worker before sharing.

6 · Seawater intake: confirm whether the sand-trap overhaul was finished; if not, ask for the money now

UNVERIFIED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
15 November 2026	Above S/6.9 million (US$2 million), per one unsourced press figure	Desalination and cooling water

The April 2026 failure stopped the reverse-osmosis unit and the FCC; Petroperú blamed seasonal sand. A July stoppage listed seawater-system maintenance with no scope or result. We could not establish completion, and a link between runoff and sediment is our inference, so you know this better than we do.

7 · Workforce: require same-day fever assessment with no anti-inflammatories, and a written heat assessment for outdoor crews, contractors included

PARTIALLY CONFIRMED FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS
30 November 2026	Site clinic time; no published figure	Workers and contractors

Fever advice follows the first item under Do not do this; Peru's clinical guide says to avoid anti-inflammatories. Same-day assessment is our advice, reasonable but without a source specific to refinery workers.* We found no specific Peruvian heat rule; Ley 29783's general duty of care applies. Give every crew the numbers 113 and 106.

8 · Water: tell EPS Grau and the municipality in writing what you can supply, and offer your structural contractor for El Acholado

PARTIALLY CONFIRMED FROM ACTIONS 1 AND 2

BY WHEN	COST TO YOU	PROTECTS
15 November 2026	None to state it. Hauling not priced	The town's water, and your standing

* Suggested formulation: check with a health worker before sharing.

A clear no is more useful than an old commitment that nobody can rely on (see Do not do this). The precedent is purchase, not production: Petroperú bought more than 90,000 gallons, about 340 m³, in January 2022, with the source unstated. Tanks of 1,000 and 2,500 m³ were promised in 2021; we could not verify completion. If they exist, they belong in the protocol.

9 · Community offers: go through the municipality, ask before proposing, and publish what you do

PARTIALLY CONFIRMED | FROM ACTIONS 6, 8, AND 9

BY WHEN	COST TO YOU	PROTECTS
20 November 2026	Fuel precedent over S/156,000 (US$45,000), 2018 prices; needs corporate approval	Drains near your gate; trust

Offer machinery on drains that affect your gate. The 2018 fuel precedent, 24,000 gallons of diesel and 2,000 of gasoline, was for post-event cleanup, and your authority to donate is unverified. Ask the municipality for the current comedor list before proposing to repeat Paremos la Olla; we found no budget, and do not know which kitchens survive. Ask APN whether its corridor means your berth, and offer no window yet. Credibility is limited: Cabo Blanco's fishers called a 2024 clean-up claim "una gran mentira", so publish evidence.

What to ask of the people in charge

- Ask EPS Grau, in writing, the current state of the El Acholado crossing and the pumps at station 1 (action 1).
- Ask the Municipality of Talara where the cistern protocol stands, and what it expects of you (action 2); and for the drainage works' contingency plan (action 6).
- Ask APN and MTC whether the corridor's mention of Talara means your terminal (action 9).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak, which is our judgment and not a modeled result. Backing lenses are shown per action. Several lenses reaching the same action is not independent proof, so each was checked against the local record. No ranked action

* Suggested formulation: check with a health worker before sharing.

has a published cost-per-death figure for a place like this.

1 · Inspect and temporarily shore the El Acholado and La Débora crossings of the 600 mm Talara water line, and stock spare pipe, by 30 November

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · HEALTH · FRONTLINE · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
EPS Grau	30 November 2026	No published price; get a contractor quote. Permitted from utility funds (Sunass)	A long water cut for more than 123,000 people

One 600 mm conveyance line, 56.3 km from El Arenal, carries water for more than 123,000 people. The Contraloría found corrosion, missing supports, and a sagging pipe at Quebrada El Acholado in 2023, and in its August 2026 visit still found missing structural elements, and maintenance left out of EPS Grau's reactive plan. We found no repair; that is absence of evidence, not proof. Permanent rehabilitation will not fit before the rains. Temporary shoring, protection of undermined supports, and spares for both ductile iron and fiberglass pipe might. A chamber of commerce account also has only two of four pumps working at station 1, with no backup power, so ask about the pumps at the same time.

* Suggested formulation: check with a health worker before sharing.

Transferability is not the issue: this line has failed here. It broke at Quebrada Débora in January 2022, leaving Talara, Negritos, Lobitos, and El Alto dry with no El Niño under way, and a bridge pipeline collapse cut water to Talara and Lobitos for weeks in 1997-98. Sunass RCD 028-2026 allows reinforcement from utility funds, but EPS Grau's share is not stated. Line length was reported as 21.7 km in 2023; this may be a segment.

2 · Sign a cistern-first water protocol between the municipality and EPS Grau, sector by sector, with a queue agreed by block, by 30 November

PARTIALLY CONFIRMED | HISTORIAN · FRONTLINE · PHILANTHROPIST · HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	30 November 2026	Standby tankers S/316,000 (US$91,000), our estimate on an unverified price	Days without safe water, and fights over trucks

Name the trucks, the filling points, and the order of sectors before the line fails. EPS Grau has named cisterns from Sullana and Máncora as fallbacks, and was sampling two wells at Quebrada Honda (undated). Trucks from Sullana use the same road the rain cuts, so the protocol needs a source on each side of the crossings. Neighborhood-council presidents helped coordinate cisterns by sector in 2022. A block-by-block queue, with neighbors carrying for those who cannot, also answers the record: in La Brea-Negritos, after 15 days without water people chased a tanker (undated report).

* Suggested formulation: check with a health worker before sharing.

Sphere's 15 liters per person per day is a humanitarian floor, not a local finding. Refinery water may be written in as a supplement only if the refinery confirms it in writing; see Do not do this.

3 · Brief every health post to refer febrile under-fives early and to assess dengue-season fevers the same day, and post a daily child-fever count, by 1 December

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Health (DIRESA, EsSalud)	1 December 2026	Staff time. Added staff S/149,000 (US$43,000) if the legal route holds	Late arrival, the failure on record in 1997-98 and 2023

The only documented cluster of El Niño deaths we found in Talara is infants, not flood victims. 18 young children died from the second week of December 1997 to mid-January 1998, mean age 9.5 months. INS called it heat-induced. A later hospital series, February to March 1998, records 6 deaths among 26 treated children and attributes it to viral encephalitis; the two counts should not be added, and the cause is disputed. Refer any febrile under-five early, especially with fever beyond 6 hours on antipyretics, convulsions, drowsiness, or breathing difficulty. Do not require that the fever have no apparent cause: only 34.6% of the fatal cases presented with fever first.* For dengue, among 40 deaths in Piura and Ica, dengue was not suspected in 40%, and 14% had self-medicated (published in Acta Tropica in 2025; we read the preprint abstract). Case fatality in 2023 was 0.90% at age 60 and over against 0.06% under 30. Numbers to post: 113 Infosalud and 106 SAMU (gob.pe).

* Suggested formulation: check with a health worker before sharing.

The credit given to community alarm and training in 1998 is the authors' uncontrolled inference. INS itself said its case definition needed validating. Facilities need confirming locally: sources show C.S. Talara II and EsSalud Hospital II, so both operators must be briefed. Talara's past share of Piura dengue is small, about 1% (DIRESA 2023 table, our sum from a garbled copy), which may also mean less immunity.

4 · Put chlorine, oral rehydration salts, and tight lids in the households most exposed to a water cut, by 15 December

PARTIALLY CONFIRMED | HEALTH · PHILANTHROPIST · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Health (DIRESA, EsSalud)	15 December 2026	S/66,000 (US$19,000), our estimate; see funding	Diarrhea and dengue during a water cut

Target the sectors that depend on trucks during a cut, not the whole town: most of Talara's water is normally piped. DIGESA's leaflet says 1 drop of household bleach (lejía) per liter, cover, shake, wait 30 minutes. Oral rehydration salts: mix the sachet in the volume on its label, usually 1 liter of boiled, cooled water, and discard after 24 hours.* Lids matter because stored water is where dengue mosquitoes breed: container covers with community clean-up gave an odds ratio of 0.22 for dengue in one study, and the review calls the evidence thin. Chlorination reduced diarrhea by about 23% (RR 0.77, low certainty). Oral rehydration reduced diarrhea deaths by 69% in measured studies in children under five.

* Suggested formulation: check with a health worker before sharing.

Evidence mostly comes from low-income settings without piped water, so the return here applies only during a cut. Press reports attribute 2 drops per liter to MINSA, which conflicts with the leaflet, and the leaflet states no bleach concentration; this needs a health worker's confirmation. Lima's diarrhea admissions rose 200% in the 1997-98 El Niño, but that was in winter and in Lima, not on this coast.

5 · Make one list per settlement of people who cannot leave alone, name who moves each one and where to, and set night quebrada watchers, by 30 November

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY · HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	30 November 2026	Volunteer time; bedding not priced	Burial and entrapment on hillsides; unnoticed heat deaths

In March 2023 two hours of rain buried hillside homes at Las Peñitas, Vista al Mar, and Fortaleza de Jesús in the Cono Norte, and water entered homes at Mario Aguirre, Nuevo Horizonte, César Vallejo, Los Ficus, and José Gálvez in the Cono Sur. In April 2023 La Débora flooded Enace (sewer collapse also contributed; a neighborhood leader put those affected at over 1,000). Walk the 2023 water paths with residents and record who is bedridden, elderly and alone, or alone with infants. Night watchers need a signal that works with no phone. Call 115 Defensa Civil or 116 Bomberos (gob.pe).* Use the same list for daily checks during SENAMHI heat notices, and refer fever or confusion: being confined to bed carried the highest heat-death risk (OR 6.44) in six case-control studies.

* Suggested formulation: check with a health worker before sharing.

Watchers and a no-phone signal are a design choice, not an evidenced intervention. Bouchama comes from Western heat waves; do not transfer Ahmedabad's 1,190 deaths averted a year, which was a multi-component plan up to 47°C, not home visits. The municipal stock holds cots and mats for 50 people, against 1,135 left homeless in the province in 2023. Close that gap; do not open shelters unplanned (see Do not do this).

6 · Get the Pariñas drainage works' contingency plan, finish the private-machinery inventory, and pre-assign machines to the open works and Avenues E and G, by 20 November

PARTIALLY CONFIRMED | EMERGENCY · SKEPTIC · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	20 November 2026	No published price	Flooding made worse by unfinished works

Talara's storm drain will not arrive in time: S/303.4 million (US$87.5 million), 3.81% physical progress, S/70 million (US$20.2 million) short in 2026, due December 2029 (July press citing Invierte.pe). Plan as though the drains do not exist, and as though open trenches will fill. The chamber of commerce recorded in August that the plan and the inventory were agreed, and that the fleet was 5 dump trucks, 2 mini-loaders, and 5 pumps, with about 12,000 sandbags. We found no later status. Avenues E and G are a mud risk during rain, so assign machines to them rather than treating them as a one-time cleanup. Clear storm drains too, but count it as restoring capacity, not as protection.

* Suggested formulation: check with a health worker before sharing.

DU 010 lets a firm do eligible services through Obras por Impuestos, with certificates charged against the municipality's own cap, so it is a faster route and not new money. Drain cleaning and pumping appear eligible in press reports; the ministry's annex was blocked. DS 149-2026-PCM lists Pariñas and all six Talara districts, but provides no new Treasury money and runs 60 days, lapsing around early January, before the summer rains, unless extended.

7 · Stage loaders at the Débora, Manuela, Pariñas, Honda, and Sícchez crossings with Provías, and stock Lobitos with water, food, medicine, and fuel, by 15 December

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · FOOD | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	15 December 2026	No published price	Days of isolation; delayed medical referral

In 2023 the Débora and Manuela quebradas cut the Panamericana at the toll booth, stranding hundreds and stopping perishables. That is Talara's own access. Lobitos' mayor blamed Sícchez, Honda, and Pariñas for its isolation, so these are separate failure points. In 1998 the Pariñas I and II bridges eroded. The municipality has only mini-loaders, and Provías handled the PE-1N here in 2023, so this needs Provías and the private inventory. Rain has already begun: El Alto took 13.5 mm on 26 September against a 0.1 mm normal.

ENFEN expects above-normal north-coast rain from November to January, with more intense and frequent rain in summer 2027. The next communiqué is 13 November. Earlier, then, is better than 15 December for any crossing that can be staged sooner.

8 · Stock the comedores for road-cut days, ask MIDIS about child meals once classes end on 27 November, and start a weekly price log, by 30 November

PARTIALLY CONFIRMED | FOOD · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	30 November 2026	Not priced: depends on the current register. Earlier lens estimates were dropped	Kitchens closing and children losing a daily meal

* Suggested formulation: check with a health worker before sharing.

This is a road, cash, and cooking problem, not yet a national shortage. Start with the municipality's PCA register: no 2026 count of comedores exists publicly (in 2022, 20 were selected and 10 awarded). Keep stock small and sized for road-cut days; markets work today, so cash works until the road fails. School meals reach 4.19 million students through the PAE, Piura's school year is reported to end on 27 November (confirm with UGEL Talara), and MIDIS's continuity plan covers supply during emergencies, not vacations. We found no plan for meals between terms. Keep a local weekly log of rice, oil, eggs, and fish, because no Talara price series was found.

Piura was cutting rice area for oversupply in May, and Midagri bought at S/3.20 per kg in August, but we found no September or October price, so we do not say a glut will last into 2027. Piura's monthly price index is reported, but it is the city and it arrives late. Catch data: Paita fishers reported 50 kg down to 2-3 kg a trip in July; no Talara figure was found, so say reduced, not collapsed.

9 · Exercise the sea route to Talara by 30 November, naming a berth, vessel, trigger, and onward trucking, or say that it does not exist

PARTIALLY CONFIRMED | FOOD · HISTORIAN · EMERGENCY | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
APN and MTC	30 November 2026	Coordination only; sailings unpriced	A corridor that exists only on paper

APN's corridor lists Talara but mentions no drill, vessel, or trucking plan. The working route, from Callao, reached Paita: its first shipment was 20 modular bridges on 4 October. A Lobitos official said in 2023 the sea would be the only way out. A trigger needs setting: we propose activating if Talara-Sullana stays closed for more than 72 hours, logged locally, since SUTRAN's alert map update frequency is unverified.

* Suggested formulation: check with a health worker before sharing.

Berth suitability is unverified. APN's list appears to class the refinery terminal as specialized liquid bulk (scrambled extraction), and swells also close this coast. An honest answer of "not possible at Talara" is still useful, because it moves the plan to the road and to stock.

10 · This week, ask the Red Cross and IFRC whether any Early Action money covers Piura, register with MIDIS for the post-event bonus, and point fishers to the Fondepes payment suspension

PARTIALLY CONFIRMED | ECONOMIST · FOOD · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Municipality of Talara	15 November 2026	Staff time	Households missed by payments that come only after damage

We could not confirm a single pre-arranged payment for Talara. A Red Cross coastal El Niño protocol naming Piura is listed both as under development and as in place, its 1,000 families span eight regions, and we found no activation. The S/2,500 million (US$721 million) household bonus is paid "when the event has effectively passed", where MIDIS detects severe problems in a province, so it moves on damage, not on a forecast. The minister has asked local authorities to help build the register, so lists are useful only through MIDIS and Pensión 65. Fishers already holding Fondepes loans may suspend installments for up to 6 months if current at 31 January 2026.

An older index-insurance design (2011 deck: pays from 24.5°C to 27.0°C on the November-December ENSO 1.2 index) would be near full payout this year, since NOAA has October Niño 1+2 at 26.12°C, 5.25°C above normal. Holding that anomaly gives a mean of about 27.5°C, which is our arithmetic. We found no live policy, sales for such a season close before it starts, and none can be bought now. It is worth one call to northern lenders in case any cover survives.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	EPS Grau	Municipality of Talara	Health (DIRESA, EsSalud)	Neighborhood councils	Refinery	ANIN	Provías	MIDIS	APN and MTC	Cáritas
1. Shore the water line	**A/R**	**I**		**I**	**C**					
2. Cistern-first protocol	**R**	**A/R**	**C**	**R**	**C**					
3. Fever referral and daily count		**C**	**A/R**	**I**	**I**					
4. Chlorine, ORS, and lids		**C**	**A**	**R**						**R**
5. Household lists and watchers		**A/R**	**C**	**R**						
6. Open works and machinery		**A/R**		**I**	**C**	**R**				
7. Plan for isolation		**A/R**		**I**	**C**		**R**			
8. Food for road-cut days		**A/R**		**C**	**I**			**C**		
9. Test the sea route		**C**			**C**				**A/R**	
10. Find the money, register		**A/R**		**R**				**C**		

What the chart shows: the Municipality of Talara is accountable for 6 of 10 rows, and its stock and machinery are thin, so it is the bottleneck. Rows 9 and 10 need a decision or a phone call, not money. EPS Grau is accountable for one row, and it is the one everything else leans on.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar, which here is our judgment, since no study gives a cost per death in this setting. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
S/215,000 (US$62,000): chlorine and clinical staff	**S/531,000 (US$153,000): adds standby cisterns**	**S/1.11 million (US$320,000): adds cash, if a recipient is found**	**The cistern price, then the cash recipient**

Arithmetic: floor 66,000 + 149,000 = S/215,000. Target 215,000 + 316,000 = S/531,000. Ceiling 531,000 + 577,000 for re-routed cash (500 households × S/1,040 food line for four × 1.11 for handling, our derivation) = S/1,108,000. **The target moved 70% in verification**, from S/1.78 million. The first ceiling of S/2.73 million meant about 185 truckloads a day, against a reported fleet of 11 to 25 trucks. Absorption before the peak is undemonstrated for every recipient here, so treat the ceiling as an upper bound and not as a request.

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Cáritas Piura with health posts, if a Talara node is confirmed	Action 4: chlorine for households exposed to a cut, January to April	S/66,000 (US$19,000)	Direct grant	US$0.31 per household-month (one 150 ml bottle), Kremer et al., Kenya and India, NBER w30835: a 30,884-household base (123,536 ÷ 4, assumed size) × 25% (assumed) × 4 months × US$0.31 × S/3.466 × 2 for delivery (assumed). The division makes this an estimate. Cited US$28 per DALY applies to under-five coupons in homes without piped water, so the return here is lower.	Partially confirmed
DIRESA Piura, for Talara facilities	Action 3: 2 doctors and 4 nurses for 4 months	S/149,000 (US$43,000)	Direct grant if legal; otherwise through DIRESA	(2 × S/12,000 + 4 × S/3,300) × 4 months, from ceiling pay advertised by MINSA (not rechecked: the source was blocked), with no employer costs. Get DIRESA's or the regional government's written confirmation that a donation can pay CAS staff. Minsa's mobile posts may overlap; Talara is not named.	Partially confirmed
EPS Grau or a contractor it names, through a pooled fund	Action 2: standby cisterns for Enace, Negritos, and Lobitos, released only on line failure	S/316,000 (US$91,000)	Pooled fund, released on line failure	123,536 × 15 L (Sphere) × 15 days = 27,796 m³ × S/45.45 per m³ × 25% (assumed share). The price comes from an undated report that could not be relocated; replace it with a quote. About 46 10-m³ loads a day, against a reported EPS Grau fleet of 11 to 25 trucks.	Partially confirmed

Dropped in verification. Larviciding 5,000 houses (S/533,000, US$153,800): the cost figures do not pair (the Machala US$30.61 is a program of patio cleanup and education that used no larvicide), and the national vector budget is S/514.86 million (US$148.5 million), over 70% executed. Cash through the Peruvian Red Cross (S/520,000, US$150,000): IFRC's Tumbes 2025 operation was run by IFRC itself because of the society's suspension, and we found no reinstatement and no Talara branch. Ask IFRC first. The state's post-event bonus may also overlap.

The funder's one thing: do not add grant money to public works. The binding limit there is execution, not cash; a former minister claims S/59.4 million (US$17.1 million) for Piura at 0% executed (not checked against MEF), and Talara holds 38 of 110 stalled prevention works. The cheapest life probably sits on one water line, and asking EPS Grau for its status costs nothing.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, or waste the window, here.

Do not give ibuprofen or other anti-inflammatories for fever in the rainy season, and do not give corticoids

CONTRAINDICATED

Among 40 dengue deaths in Piura and Ica, 14% had self-medicated with anti-inflammatories or corticoids, and Peru's clinical guide says to avoid them. MINSA's page also warns against metamizole and ivermectin. 5 of Piura's 9 reported dengue deaths to week 40 this year were minors (7 confirmed, 2 under investigation).

Instead: adults may take paracetamol 500 mg every 6 to 8 hours, 3 g a day at most; children only on prescription. Go to a health post the same day, or call 113 or 106.*

Do not plan on refinery water as Talara's backup supply

CONTRAINDICATED

A 2,000 m³ a day commitment was reported in February 2023, but we found no evidence of delivery. A Congress record seen only as a search excerpt has Petroperú's general manager saying the water is for boilers, not potable, and not available without state funding. It would be 91% of the 2,200 m³ a day listed capacity. The plant was disabled in December 2024, failed in April 2026, and the refinery stopped on 1 November.

* Suggested formulation: check with a health worker before sharing.

Instead: write the protocol around cisterns and wells, and add refinery water only when the refinery confirms the volume and its potability in writing.

Do not tell people simply to store more water

CONTRAINDICATED

Open drums turn a water cut into a dengue outbreak. In the 2023 Piura epidemic, water rationed to 30 minutes a day forced storage in containers that became breeding sites. Past low dengue here may also mean low immunity (our inference).

Instead: tight lids, weekly scrubbing, and community clean-up. The combination had the strongest result in the review, from one study.

Do not make fogging the visible main dengue response

CONTRAINDICATED

There are no randomized trials of fogging against dengue, which is absence of evidence and not proof it fails. Mosquitoes in Piura and Castilla in 2014 showed pyrethroid resistance (deltamethrin 74.7% and permethrin 12.9% mortality), twelve years ago and not in Talara.

Instead: reserve it for targeted outbreak use, and put the effort into containers and screens.

Do not count channel clearing as protection against large flows

CONTRAINDICATED

It restores capacity and no more. The CAF review of 1997-98 credits cleaning, bundled with other works, with avoiding serious river floods on many occasions, without quantifying it, and records that design flows were exceeded and that Piura's drains ran full and still were not enough. One engineer called the S/346 million (US$100 million) spent on the Piura river in 2017 earth removal; that is an opinion on a river, not a study of Talara's drains.

* Suggested formulation: check with a health worker before sharing.

Instead: clear the drains, but also map where water went in 2023, and agree exits and meeting points on high ground.

Do not open a shelter without land title, services, and an exit plan

CONTRAINDICATED

Cura Mori's 2017 Km 980 shelters became settlements with no water, power, schools, or clinic, on privately titled land, and residents faced eviction in 2023. It is one case, in another province.

Instead: agree the land, the services, and an end date before anyone moves in, and do not rebuild in a quebrada bed.

Do not plan mass doxycycline against leptospirosis

CONTRAINDICATED

The review finds insufficient evidence for mass use. The best local-type result, 95.6% among people with wounds, was non-randomized (600 versus 41). Piura's 2,682 cases and 8 deaths are undated, only 506 are confirmed, and Talara is not named.

Instead: ask clinicians about short-course use for high-exposure adults, such as cleanup crews with wounds. Brazilian guidance excludes pregnant and breastfeeding women.*

Do not treat a Fondepes loan as an income bridge for fishers

CONTRAINDICATED

It is credit, for a crew with less catch. Press reports 1% a year, up to 15 UIT, S/5.99 million (US$1.73 million) nationally, applications by 15 December; fondepes.gob.pe could not be reached. The minister said in September that loans would come in kind, as engines and gear.

Instead: use it for gear, if at all. Ask first about the installment suspension (action 10), and get crews onto the MIDIS register.

* Suggested formulation: check with a health worker before sharing.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1983	Talara: "Catastrophic floods" in Talara.	No local toll found	Nothing found
1997-98	Talara: 18 young children die, mean age 9.5 months; INS calls it heat-induced. Water and sewer gaps noted.	18 children (INS, hospital record)	Health posts trained, by the later study's account
1998	Talara: Hospital series: 26 children treated, attributed to viral encephalitis. Separate window; do not add to the above.	6 of 26 treated	Earlier referrals credited, without a control
1998	Pariñas, Negritos, Lobitos: Pariñas I and II bridges eroded; Negritos-Talara road eroded; bridge pipeline collapse cuts water to Talara and Lobitos for weeks.	No local toll found	Talara still depends on one line today
2017	Piura Region: Coastal El Niño. Region: dengue about 35,000 cases and 33 deaths (interview figure). El Acholado crossing damaged, per local press. Province-wide isolation could not be verified.	No Talara toll found	Reconstruction promised; storm drain still 3.81% built in 2026
2022	Talara, Negritos, Lobitos, El Alto: Water line breaks at Quebrada Débora with no El Niño under way. Petroperú buys more than 90,000 gallons.	None reported	Sunass opens enforcement, citing missed repair deadlines
2023	Talara province: Cyclone Yaku period. Province: 1,135 homeless, 7,271 affected, 59 homes destroyed. Pariñas: 725, 2,821, 38 destroyed, 161 uninhabitable. Panamericana cut at the toll booth.	No deaths recorded (INDECI)	Contraloría warns of line collapse
2026	Máncora, Los Órganos: At least 200 homes collapse in Máncora (preliminary, local authorities); Panamericana blocked.	None found	Contraloría warns again in August

The recurrence nobody has noticed: everyone watches the sky, but the same pipe keeps failing, including in a year with no El Niño. A town in a desert, in its wettest months, depends on trucks that use the roads the same rain cuts. We found no Talara death toll for 1983, 2017, or 2023's rains beyond what is listed, so the rain's true toll here is unknown.

* Suggested formulation: check with a health worker before sharing.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Rain is the killer** versus **heat may be, unseen**. Most lenses put rain first. The Public Health lens says that compares counted deaths with modeled ones. At about 7% of the region's people, Talara's share of the 360 forecast heat deaths would be about 25, which is our arithmetic and not a Climate Impact Lab finding, and the sea breeze probably lowers it. The same 7% against about 1% of Piura's dengue cases gives 1 to 2 of 2023's 130 regional deaths. A count of death certificates by cause would help settle it.
- **The 1997-98 infant deaths were heat** (INS) versus **viral encephalitis** (Del Águila and Briceño). It decides whether cooling or infection control protects infants next time. Referring early helps in both cases, so action 3 does not wait for the answer.
- **Clear the channels** (Historian, Emergency Manager) versus **clearing is theater** (Skeptic). Verification narrowed this: storm-drain cleaning restores capacity and is worth doing, but no head-to-head study exists. The distinction between drains and rivers is our inference.
- **Buy rice now, in the glut** (Food lens) versus **cash works until the road fails** (Food verifier). No autumn price was found. We kept a small stock sized for road-cut days.

WHAT NOBODY COVERED — BLIND SPOT

The electric grid. No lens looked at it, yet a national grid event on 1 November cut the line between Chimbote and Trujillo and stopped the refinery, and EPS Grau's pump station reportedly has no backup power. Power loss could take the water line down without any quebrada touching it. Also unstudied: migrant households, Talara's own housing and age structure, and who actually eats last when the road is cut.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are at province level: all of Piura Region, not Talara. The heat figures count heat only; drought, flood, fire, smoke, and crop failure are additional and are carried here by the historical record rather than by a published number.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as given: Talara, Peru, read as the city of Talara in Pariñas district, capital of Talara Province, with the province where sources report it. Mortality figures exist only at region level.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Piura Region, 110 additional heat deaths June–September 2026, 360 forecast October 2026 – March 2027 (October 41, November 62, December 94, January 57, February 59, March 43), 460 in total; Peru, 3,200; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted in an action's favor. Contradictions were kept, not resolved by dropping a side. Actions were ranked by our judgment of lives saved per unit of effort before the peak, and the funding placements were reconciled against the ranked list.
- **Verification.** Seven verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. For you steps were judged for one refinery manager.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organization named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Its "Download Inventory" link leads to a registration form, and the browse page loads its data in the browser. No action here should be read as having no precedent; none was checked.
- **Money.** Costs are shown in Peruvian soles with US dollars in parentheses, at S/3.466 = US$1 (Banco Central de Reserva del Perú, banking-system sell rate, 2 November 2026). Figures published in dollars are converted at that rate, including older amounts, which are given for

* Suggested formulation: check with a health worker before sharing.

scale only.

- **Reader.** The requester described themselves as a refinery manager. The "For you" steps were taken from the verified actions and from what the panel found about the refinery here; any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the municipality and its partners.
- **Health wording.** Passages where a wording error could hurt someone (fever and when to seek care, medicines and doses, water chlorination, oral rehydration salts, emergency calls, and antibiotics) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first.
- **Rules.** No cost is derived by dividing a published total without showing the division and calling it an estimate, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Several Peruvian government and press pages blocked automated reading, so some figures rest on search excerpts or press quotations, and we say so where it matters. The Congress record on boiler water, the current state of the refinery's plants, and the status of all works after August 2026 could not be confirmed. A report of homes affected by rain in Talara in early November could not be opened and is left out. Dates in the For you section are our proposals.

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* Suggested formulation: check with a health worker before sharing.

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Exposure data · Climate Impact Lab, El Niño Heat Mortality Tracker data, v261026 · Impact Map · Adaptation Inventory

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